DALAS
PROJECT

PREDICTING THE PRICE OF A RESTAURANT DISH

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Introduction

The price of a dish in a restaurant depends on many different factors. These include the cost of ingredients, the type of cuisine, the location of the restaurant, and its popularity among customers. In a city like Paris where restaurants are numerous and very diverse it's common to observe large price differences between dishes that may look similar.
However, the reasons behind these differences are not always clear. Today online food platforms and restaurant websites provide public access to a large amount of information about restaurants and their menus. Prices, dish descriptions, cuisine types, customer ratings and locations are easily available online. This data makes it possible to study restaurant pricing using a data-driven approach based on real observations rather than assumptions.

The goal of this project is to predict the price of a restaurant dish using publicly available data collected from online platforms. We focus on restaurants located in Paris and consider both dish-level information (such as the name and description of the dish) and restaurant-level characteristics (such as cuisine type, location, and customer ratings). The problem is formulated as a supervised regression task, where the target variable is the price of the dish in euros.

In addition to building predictive models, this project aims to better understand which features have the strongest influence on dish prices and to identify situations where the models perform well or poorly. Since the data is collected through web scraping it may contain missing values, noise, and biases, which are taken into account in the analysis.
Overall, this project follows a classical data science workflow, including data collection, preprocessing, modeling, and evaluation of the results.

Data and Context

To address the problem of predicting dish prices, we rely on publicly available data collected from online food delivery platforms. These platforms provide information at both the dish level and the restaurant level, which makes it possible to analyze how different observable characteristics relate to pricing.

a) Data sources and Web-Scraping

The main data source used in this project is Uber Eats. This platform provides access to restaurant menus, including dish names, textual descriptions, prices, cuisine categories, restaurant ratings, and the number of customer reviews. This information was collected through web scraping techniques.

Web scraping was performed by automatically extracting information from publicly accessible restaurant pages. The scraping process focused on collecting structured data such as dish prices and categories, as well as unstructured textual data such as dish descriptions. Due to technical limitations, including dynamic content loading and anti-bot protection mechanisms, the data collection process could not retrieve the complete menu for every restaurant.

As a result, the dataset represents a representative view of restaurant offerings. Other platforms such as Deliveroo, TheFork, and TripAdvisor were initially considered.
However, we observed that the information available on Deliveroo largely overlapped with Uber Eats, while scraping TheFork and TripAdvisor proved difficult due to access restrictions. For these reasons, the analysis was mainly based on data collected from Uber Eats.

All data used in this project comes from publicly accessible web pages and does not include any personal or private information.

b) Dataset description

The final dataset is structured as a single tabular file in which each row corresponds to one dish offered by a restaurant located in Paris. The dataset contains approximately 14,000 observations and around 12 variables. The target variable is the dish price, expressed in euros. Each row of the dataset corresponds to one dish sold by a restaurant in Paris.

The dataset contains information about the dish itself and about the restaurant :

- **Nom restaurant** : this variable is an identifier and is not used directly for prediction.
- **Adresse** : this variable contains detailed location information.
- **Code postal** : since the study focuses on Paris, this variable is used to derive the arrondissement (district).
- **Nom plat** : this is a textual variable that provides information about the type of dish.
- **Prix : this variable is the price of the dish** as displayed on the platform. This variable is the target variable of the study. In the raw data it appears in different formats and requires cleaning.
- **Description :** : this variable is the **textual description of the dish. It** may include information about ingredients or preparation methods.
- **Section : shows the menu section** in which the dish appears (for example pasta, burgers, sushi) : this is a categorical variable.
- **Note : is the average customer rating of the restaurant**. This variable is numerical but initially stored as text (for example “4,5”).
- **Avis : represent the number of customer reviews** available for the restaurant. This variable is stored as text (for example “(440+)”).
- **Catégorie fixée : represent the main cuisine category of the dish. It is** assigned to the restaurant (for example “Pâtes – IT”), this is a categorical variable.
- **Catégories : displays more detailed cuisine labels** associated with the restaurant. This variable has high cardinality and heterogeneous labels.
- **Prix catégorie : represent the price range indicator** displayed on the platform (for example €, €€, €€€). This variable is categorical and ordinal by nature.

c) Context and scope of the study

This study focuses exclusively on restaurants located in Paris and on dishes available through online delivery platforms. As a consequence the dataset mainly represents restaurants that offer delivery services and may under-represent certain types of establishments, such as high-end or traditional restaurants that are less active online. Despite these limitations, the dataset provides a large and diverse sample of dishes and restaurants across different cuisine types and price ranges. This makes it suitable for analyzing general pricing patterns and for building predictive models based on publicly available information.

Data quality verifications

The dataset used in this project was collected through web scraping from online food delivery platforms. As expected with scraped data the raw dataset contained several quality issues that required careful preprocessing before any modeling could be performed. These issues include missing values, duplicated entries, heterogeneous formats, and platform-related biases.

a) Data quality issues

Several data quality problems were identified in the initial dataset. First, missing values were present in multiple variables especially for customer ratings, number of reviews, and dish descriptions. Not all restaurants provide the same level of information on the platform which leads to incomplete observations. Second, the dataset contained duplicate entries, mainly due to restaurants or dishes appearing multiple times during the scraping process or across different extraction sessions. Some dishes were listed several times with identical or very similar information. Third, a strong heterogeneity in labels and formats was observed. For example, dish categories and cuisine types were not standardized and could appear under different names for similar items. In addition, numerical variables such as prices, ratings, and number of reviews were stored in inconsistent formats.

A specific issue was observed with the price variable. During the extraction process and subsequent handling of the data in Excel, some price values were incorrectly interpreted as dates due to automatic formatting. This required manual inspection and reformatting to restore prices to their correct numerical representation.

b) Data cleaning and processing

Data cleaning was mainly carried out using Power Query in Excel, which allowed for an efficient and transparent treatment of the dataset. The cleaning process was performed variable by variable to ensure consistency and correctness. Duplicate rows were identified and removed directly on the platform. Numerical variables such as prices, customer ratings, and number of reviews were reformatted to their appropriate types. For the number of reviews, non-numerical characters such as parentheses were removed before conversion to integers. Location-related variables were also simplified. Full address and postal code information was reduced to district-level information in order to avoid redundancy and reduce noise.

Textual variables such as dish names and descriptions were cleaned to remove obvious formatting errors introduced during scraping. This step helped ensure that textual data could later be processed correctly during the modeling phase.

c) Biases

A platform bias is inherent to the data collection process. The dataset mainly represents restaurants that are active on food delivery platforms, which excludes many traditional or high-end restaurants that do not rely on online delivery services. As a result, the dataset is biased toward mid-range and fast-food establishments. There is also a visibility bias, as more popular restaurants tend to have more reviews and more complete information, while less popular restaurants are under-represented. Simplifications introduced during preprocessing, such as category restriction or aggregation of location information, may reduce variability in the data. However, these choices were necessary to improve data quality and reduce noise, and their impact is taken into account when interpreting the results.

d) Feature engineering and preprocessing

Feature engineering was used to make the data suitable for modeling and to improve predictive performance. In addition to numerical and categorical variables, textual information from dish names, dish descriptions, and dish sections was included. Textual features were transformed using TF-IDF vectorization. To limit dimensionality, only the most frequent terms were kept. For dish names and dish sections, the 300 most frequent features were retained, while 500 features were kept for dish descriptions, which contain more detailed information.

This approach makes it possible to incorporate textual information into the models while controlling the size of the feature space.

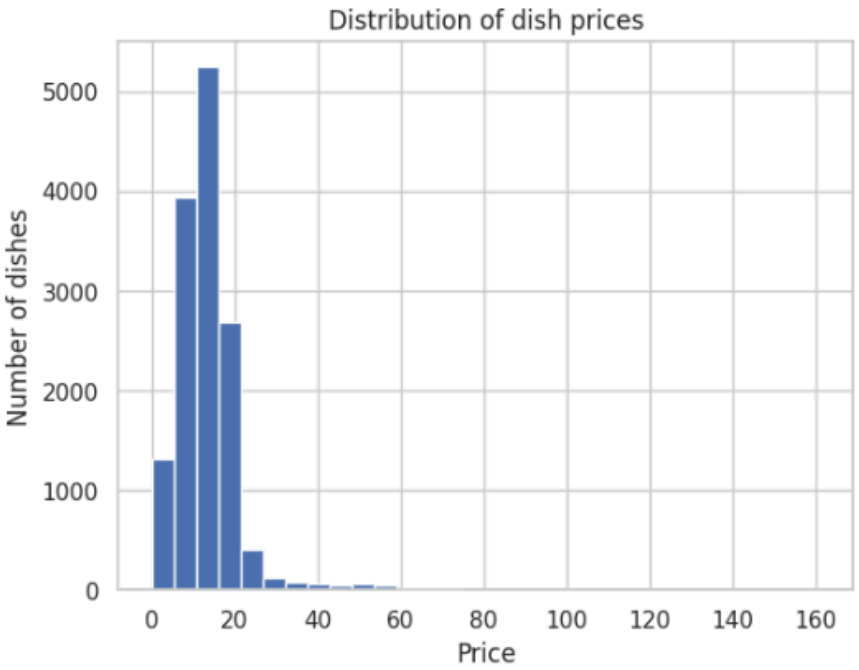
e) Models and methodology

Two regression models were trained and evaluated in this study. A Linear Regression model was first used as a baseline, as it is simple and easy to interpret. This model assumes a linear relationship between the input variables and the dish price. A Random Forest Regressor was then used to capture more complex patterns in the data, including non-linear relationships and interactions between variables. The performance of both models was evaluated using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score.

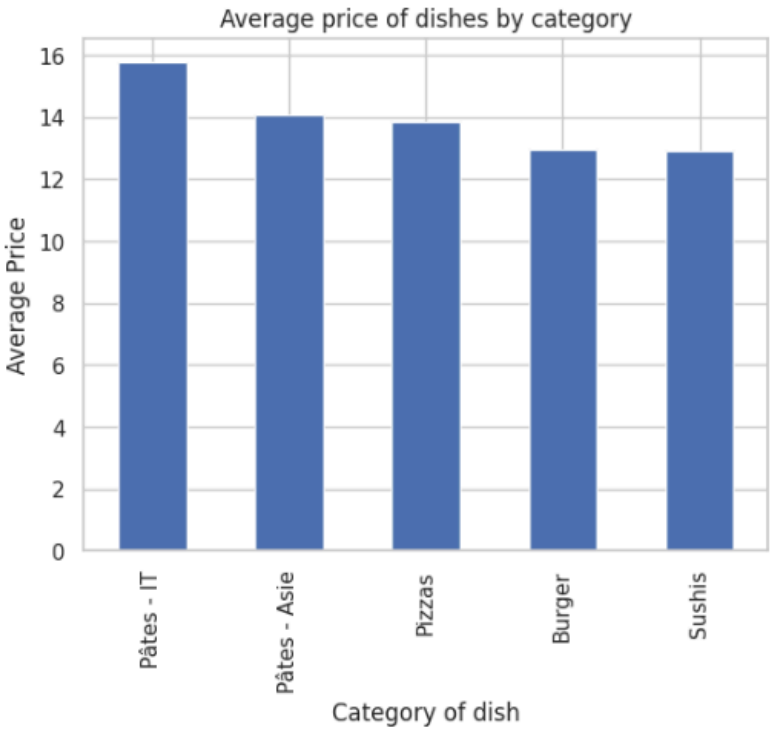
The data was split into 80% for training and 20% for testing in order to evaluate the models on unseen data.

Results and analysis

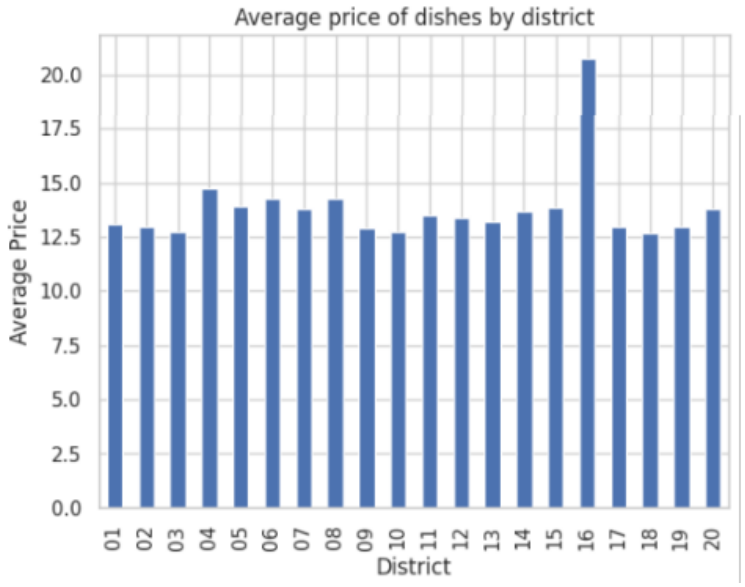
a) Descriptive analysis

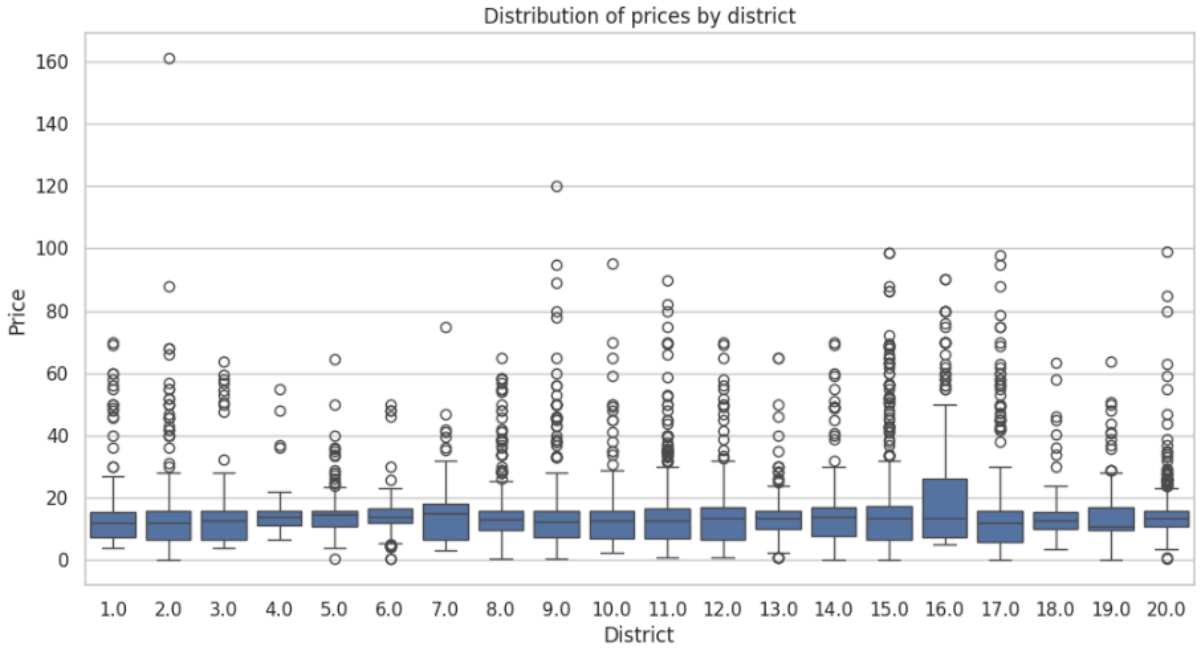


The distribution of dish prices is highly right-skewed. Most dishes are priced between approximately 8 and 20 euros. This shows that the dishes collected have similar prices.



When comparing prices across dish categories, noticeable differences appear. Italian pasta dishes have the highest average prices, followed by Asian pasta and pizzas. Burgers and sushi tend to have slightly lower average prices. However, the differences between categories remain moderate, suggesting that category alone cannot fully explain price variability.





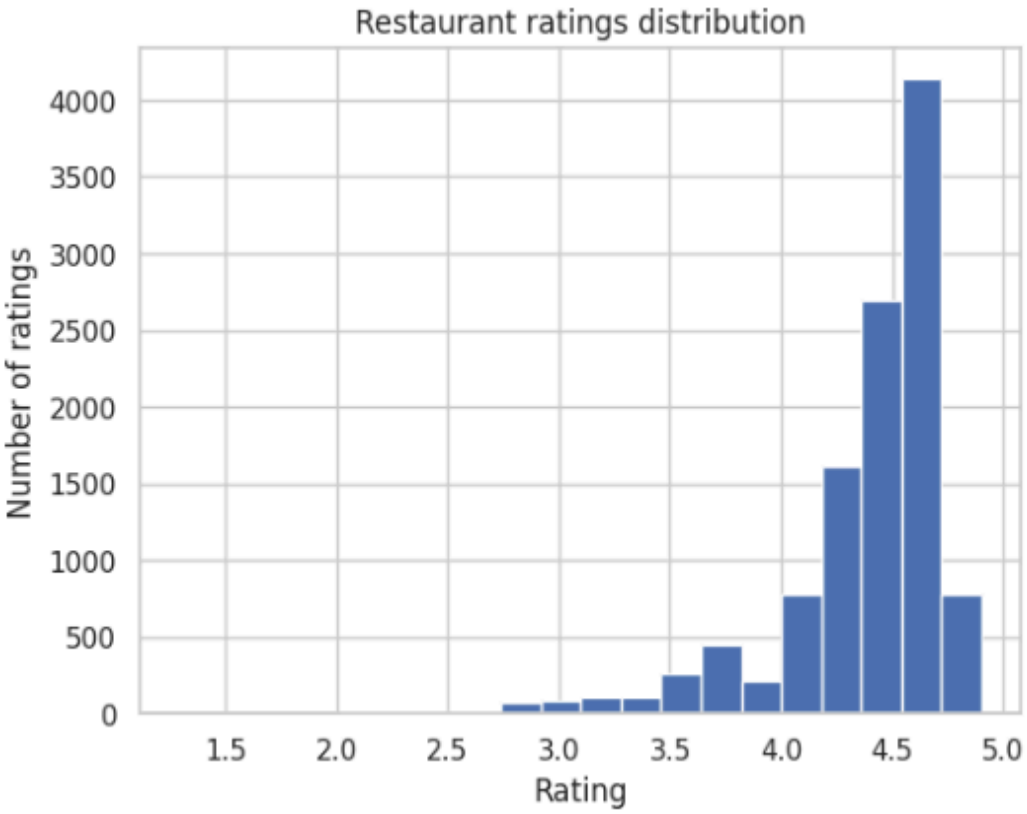


The analysis by district shows that average prices vary across districts, with some higher values observed in 16th district for example. Nevertheless, price dispersion within each arrondissement is large indicating that location has an influence on pricing but is not a determining factor by itself.

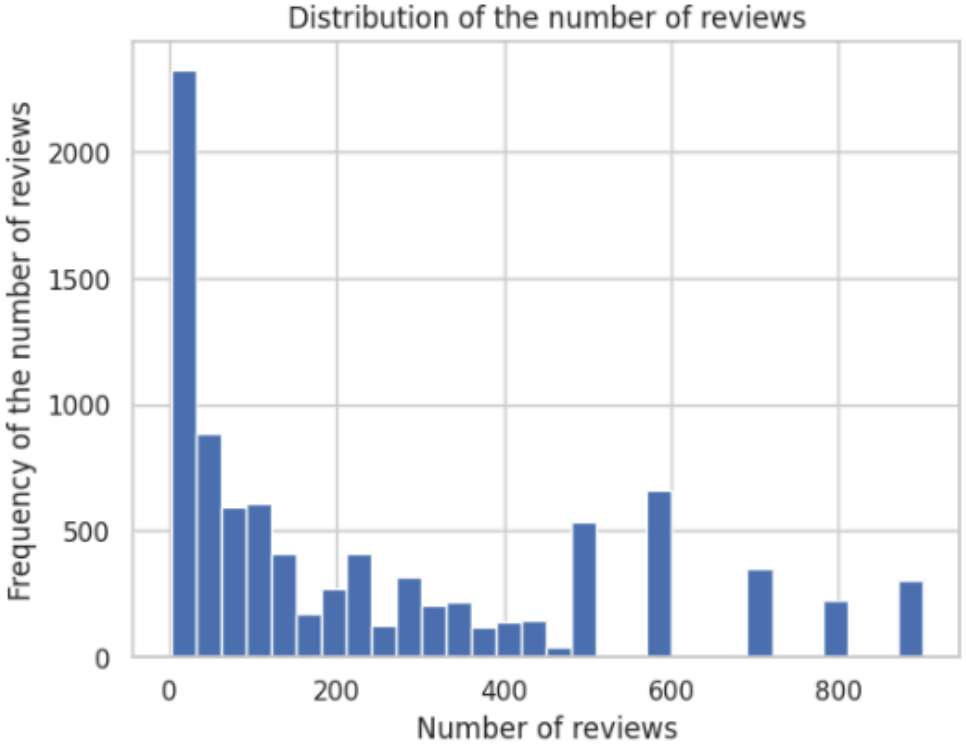


The two categories have similar mean and median. However, there are 9,400 restaurants classified as €€ compared to only 1,200 restaurants classified as €. This similarity is undoubtedly linked to the underrepresentation of €€ restaurants in the dataset.

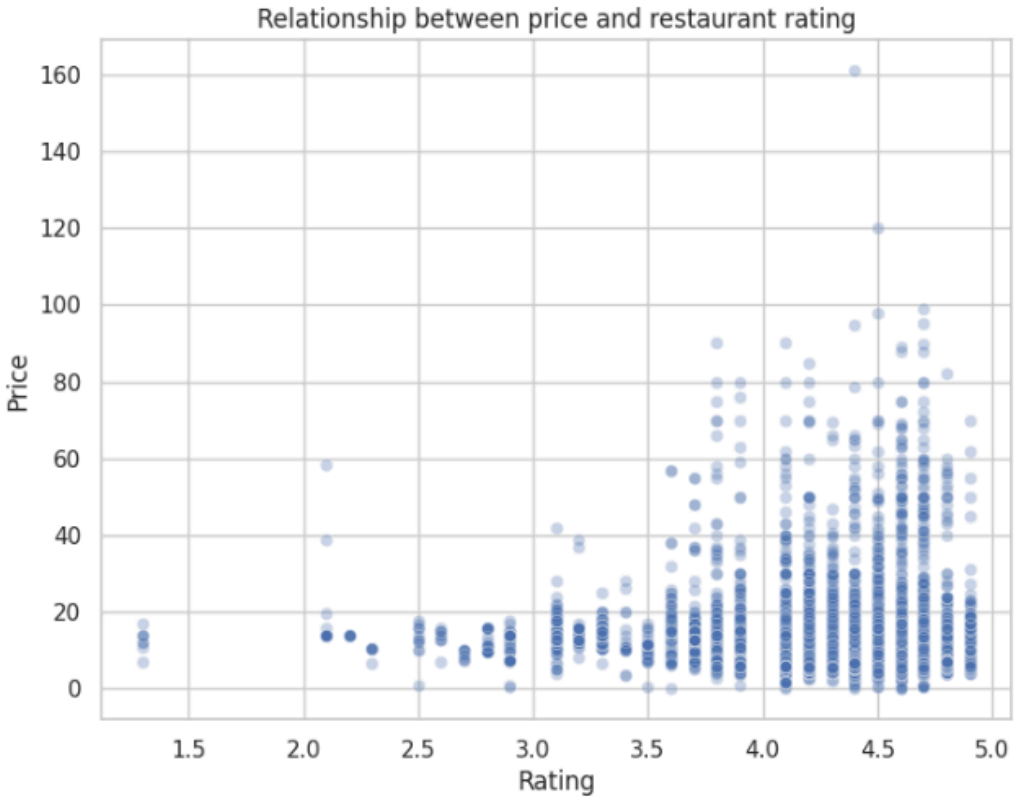
b) Ratings, popularity and bivariate analysis



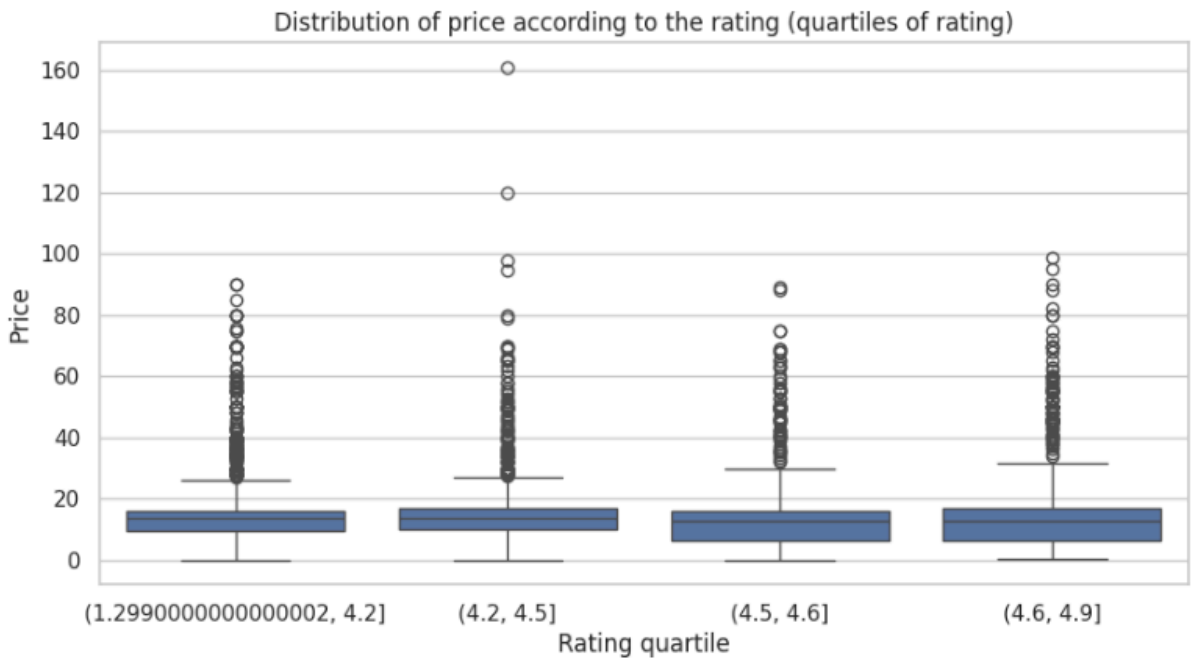
Restaurant ratings are strongly concentrated between 4.0 and 5.0, which limits their discriminative power. This can be explained by the fact that users tend to review restaurants they like rather than the ones they don't like.



The number of reviews follows a highly skewed distribution, with a small number of restaurants accumulating a very large number of reviews. The gaps can be caused by the way reviews are implemented in Uber Eats. Beyond a certain number of reviews, the number of reviews are represented by intervalles. For instance 440+, 1000+, 1500+.



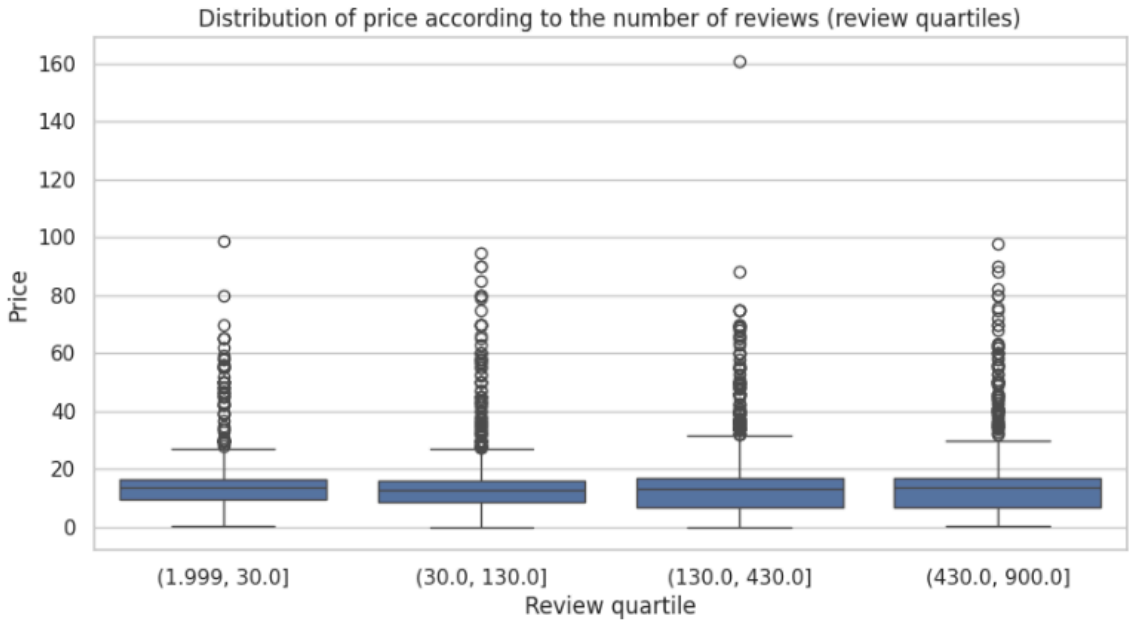
Bivariate analysis shows that the relationship between price and rating is weak. This shows that customers don’t take the price in account when doing a review.



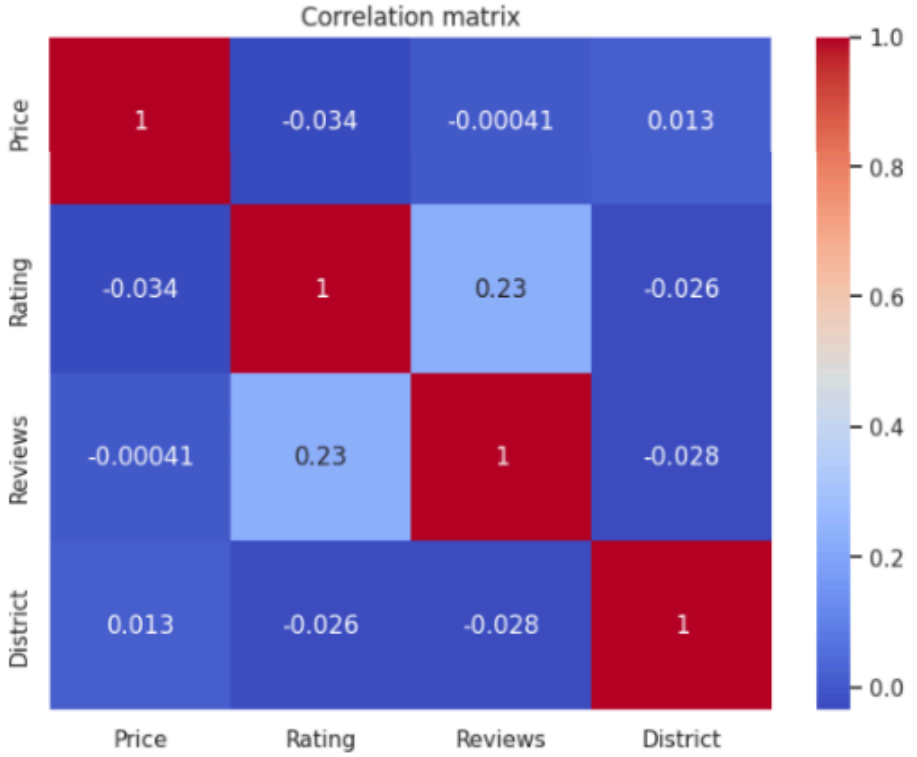
Higher-rated restaurants do not systematically charge higher prices, although expensive dishes tend to appear more frequently among restaurants with high ratings.



This scatter plot shows no clear relationship between dish price and the number of reviews. Most dishes are priced below 30 euros, regardless of how many reviews the restaurant has. This suggests that restaurant popularity has a limited influence on dish prices.



Similarly, the relationship between price and the number of reviews is limited, suggesting that restaurant popularity alone does not strongly drive pricing.



The correlation matrix shows that price has only weak linear correlations with most explanatory variables, including rating and number of reviews. This indicates that

simple linear relationships are insufficient to explain pricing behavior and motivates the use of non-linear models such as Random Forests.

c) ML results

The Random Forest model consistently outperformed the Linear Regression baseline across all evaluation metrics, indicating the presence of non-linear patterns in the data. The models performed particularly well for dish categories like sushi, pizzas and Italian pasta.

```
Size train : (11297, 1632)
Size test  : (2825, 1632)
Performance of the Linear Regression
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- MAE : 2.98 €
- RMSE : 4.99 €
- R^2 : 0.675

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Performance of the Random Forest
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- MAE : 2.72 €
- RMSE : 4.61 €
- R^2 : 0.722

Mean Absolute Error (MAE):

➔ Random Forest : 2.72 €
➔ Linear Regression : 2.98 €

Random Forest reduces the mean error by approximately 0.26 € per dish. This means the predicted price is closer to the actual price for the majority of dishes.

Root Mean Squared Error (MSE):

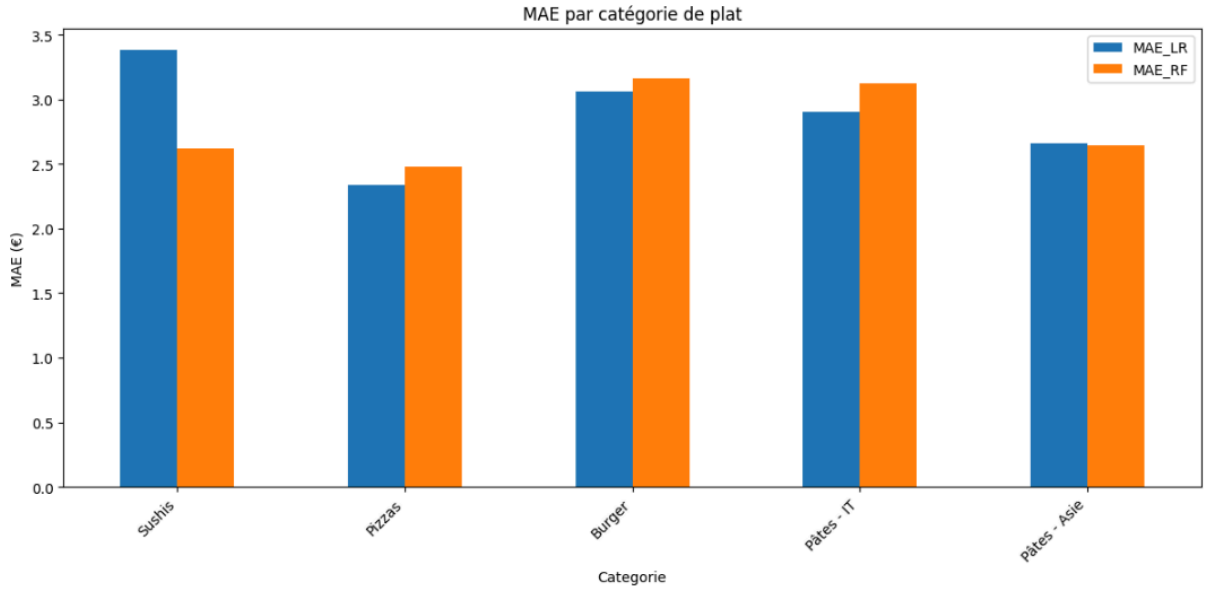
➔ Random Forest : 4.61 €
➔ Linear Regression : 4.99 €

RMSE penalizes large errors more heavily. The observed decrease indicates that Random Forest handles extreme prices and limits significant errors, particularly for expensive or atypical dishes.

Coefficient of determination (R^2):

➔ Random Forest : 0.722
➔ Linear Regression : 0.675

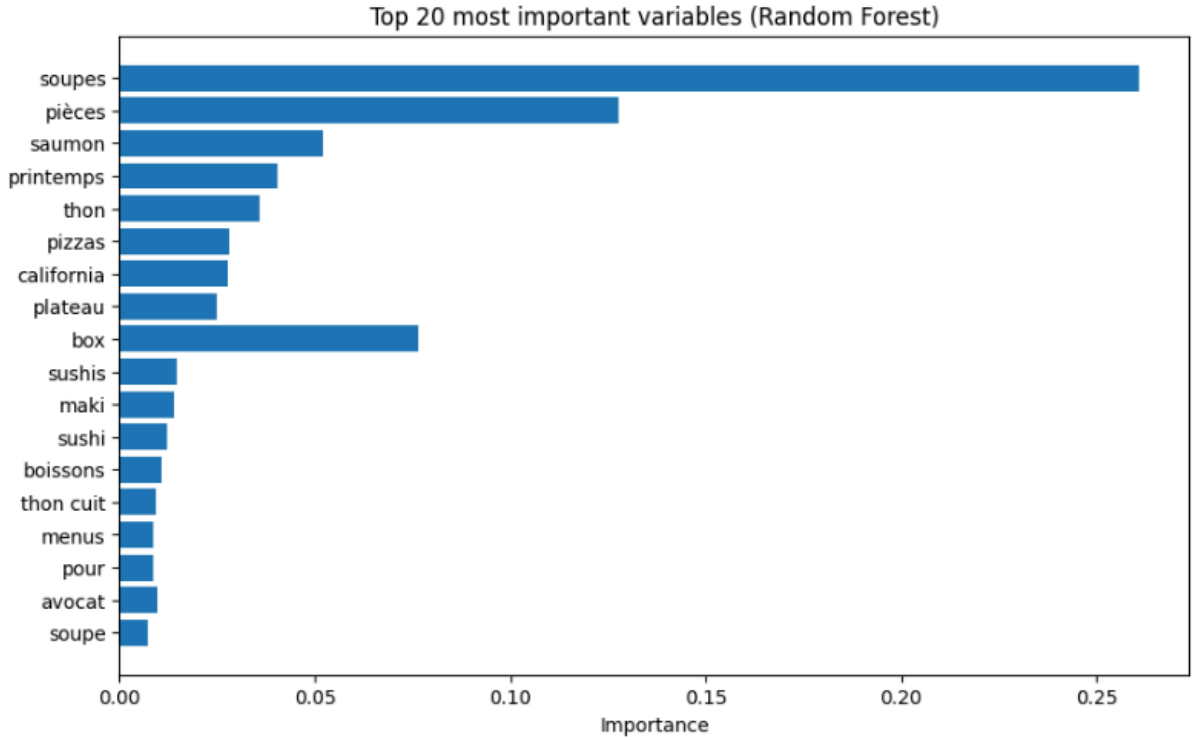
Random Forest explains approximately 72.2% of the price variance, compared to 67.5% for linear regression. This confirms that the relationships between features and price are non-linear.



While the Random Forest model achieves higher overall R^2, the linear regression exhibits lower MAE for most dish categories, suggesting that price variations within these categories are mostly linear. The exception is the sushi category where there seems to be stronger non-linear interactions. This suggests that the price for this category is strongly correlated to the number of pieces in the dish or if the dish is a box/platter of sushi.

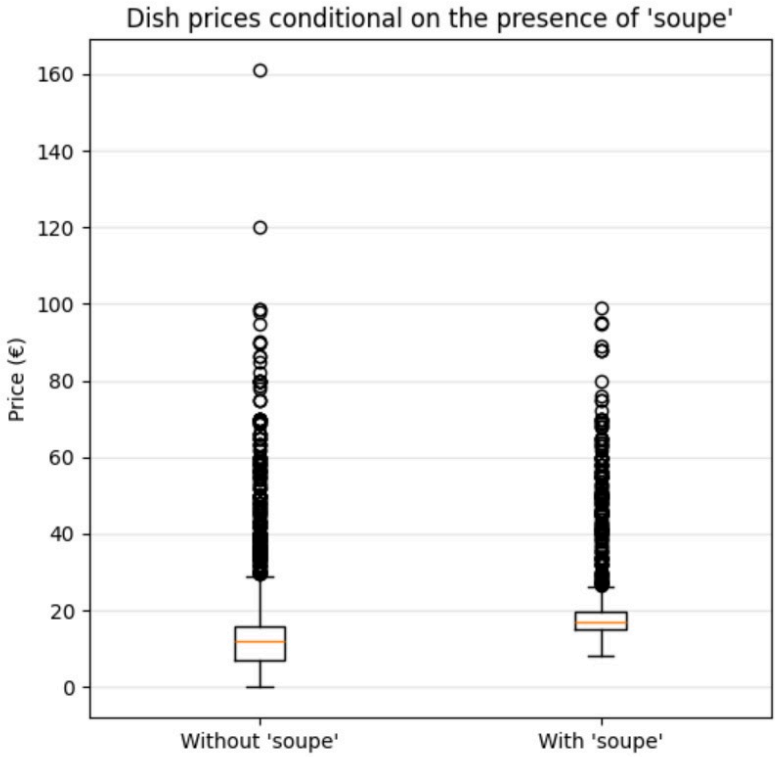
Interpretation

The feature importance analysis highlights the variables that contribute the most to the Random Forest model's predictions.



As shown in the figure, the most influential features are mainly related to dish type and textual information extracted from dish names and descriptions. Several TF-IDF features appear among the most important variables, which indicates that textual information plays a significant role in price prediction. Many of the most important words are related to sushi dishes, such as "pièces", "saumon", "thon", "california", "plateau", "box", "sushis", and "maki". This result is consistent with the previous analysis, where the Random Forest model achieved better performance for the sushi category. It suggests that pricing patterns within this category are more complex and benefit from the use of detailed textual information.

The word "soupes" also appears as a highly important feature. This can be explained by the relatively low price variability observed for soups compared to other dish types, which makes their prices easier for the model to predict.



Finally, terms such as “pièces”, “box”, “plateau”, and “menu” are also identified as important. These words are generally associated with dishes composed of multiple items, which are usually sold at higher prices. This indicates that the model is able to capture information related to dish composition through textual features.

Overall, this analysis confirms that the Random Forest model relies primarily on dish-related and text-based features, while restaurant-level variables play a secondary role in price prediction. It is important to note that feature importance values in Random Forest models do not imply causal relationships. They only reflect how useful a variable is for prediction within the model. In addition, feature importance values may vary depending on the training data and model initialization. Therefore, the interpretation focuses on general trends rather than exact importance rankings.

Conclusion and limits

The goal of this project was to predict the price of restaurant dishes in Paris using publicly available data collected from online food delivery platforms. The problem was approached as a supervised regression task, using both dish-level information and restaurant-related features. The exploratory analysis showed that dish prices vary a lot, especially depending on the type of dish. Location has an influence on prices but it is not enough on its own to explain large price differences. Restaurant ratings and the number of reviews are only weakly related to prices which suggests that popularity is not a main factor in pricing. In terms of modeling, the Random Forest model performs better than the linear regression baseline on all evaluation metrics. This shows that the relationship between the features and the dish price is not purely linear. Textual features extracted from dish names and descriptions play an important role in improving the predictions.

However, this study has several limitations. The data was collected through web scraping and therefore contains missing values and noise. In addition, the dataset mainly includes restaurants that are active on delivery platforms, which creates a bias and limits how representative the results are. Some important information, such as portion size or ingredient quality, is not directly available and can only be approximated through text. Finally, the interpretation of the model results should be done carefully. Feature importance does not imply causality, and the results may vary depending on the data used to train the model.

Overall, this project shows that publicly available data can be used to model restaurant pricing, while also highlighting the limits of such an approach.